MCQ QUESTIONS

Cell Biology

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Multiple-Choice Quiz - Cell Biology

Target age: 13-17 years

Q1. [Difficulty: Easy]
Which organelle is known as the powerhouse of the cell?

A) Ribosome

B) Mitochondria

C) Nucleus

D) Endoplasmic Reticulum

✔ **Correct Answer: B**

Explanation: Mitochondria generate most of the chemical energy needed to power the biochemical reactions of the cell.

Q2. [Difficulty: Medium]
What is the primary function of the cell membrane?

A) Synthesizing proteins

B) Controlling what enters and leaves the cell

C) Replicating DNA

D) Packaging enzymes

✔ **Correct Answer: B**

Explanation: The selectively permeable lipid bilayer controls the influx and efflux of ions and molecules.

Q3. [Difficulty: Hard]
Which of the following processes occurs entirely across the inner mitochondrial membrane?

A) Glycolysis

B) Krebs Cycle

C) Electron Transport Chain and ATP synthesis

D) Fermentation

✔ **Correct Answer: C**

Explanation: The ETC proteins and ATP synthase complexes are embedded in the inner mitochondrial cristae.